Thermodynamic description of the Cu–Pb system: a synthetic benchmark document

ORCHID synthetic benchmark (generated; no human authors)

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Abstract

This synthetic document describes the Cu–Pb binary system as calculated with pycalphad from the SGTE dataset CuPb-86Hay-LB.tdb; every figure, table and stated value is generated, and the exact values are kept in a separate truth key. The highest invariant reaction, L1 → (Cu) + L2, lies at 956.8 °C.

1. Introduction

The Cu–Pb system belongs to the 20 binaries of the ORCHID benchmark, chosen to cover diagram types such as miscibility gap, monotectic. The thermodynamic description is the SGTE collection entry (Hallstedt, 2025), which reproduces the assessment of F.H. Hayes, H.L. Lukas, G. Effenberg, G. Petzow, Z. Metallkd., 77, 749-54(1986). The calculations use pycalphad 0.11.2 at 1 atm.

For Cu, the liquidus meets the axis at 1084.6 °C.

2. Phase diagram

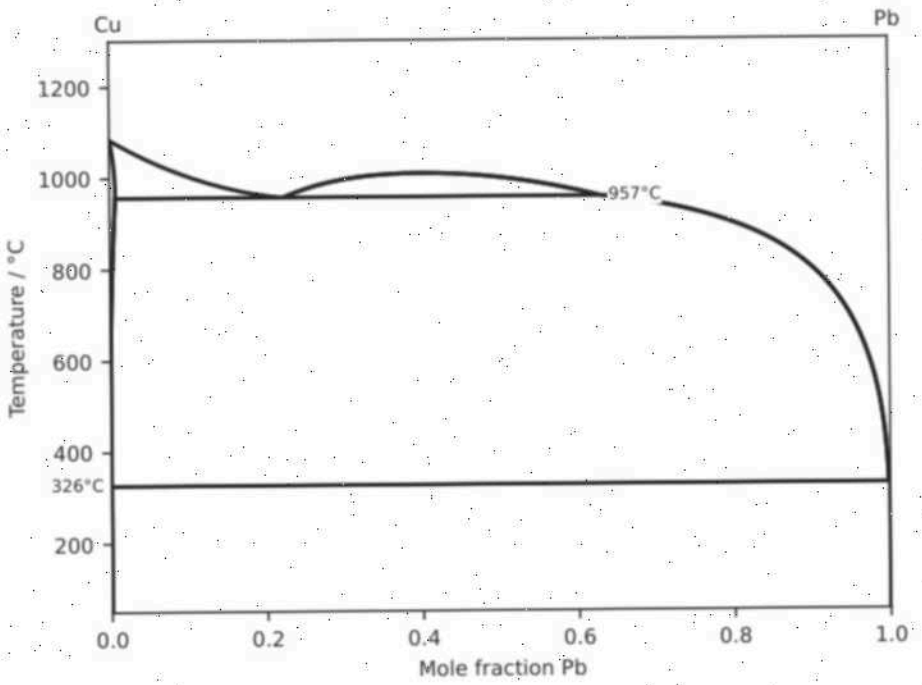


Fig. 1. Calculated Cu–Pb phase diagram.

At 956.8 °C the monotectic L1 → (Cu) + L2 takes place; the L1 phase then contains 22.2 at.%. The eutectic reaction L → (Cu) + (Pb) is calculated at 326.3 °C, with L at 99.8 at.%.

Table 1. Invariant reactions in the Cu–Pb system (calculated).

Reaction	Type	T / °C	Compositions (x(Pb))		
			phase 1	phase 2	phase 3
L1 → (Cu) + L2	monotectic	956.8	0.222	0.007	0.631
L → (Cu) + (Pb)	eutectic	326.3	0.998	0.000	1.000

3. Thermodynamic properties

At 1450 K the integral mixing enthalpy of the liquid has its maximum, 6.81 kJ/mol, at 48.0 at.% Pb. The calculated enthalpy of formation of HCP_A3 at 298.15 K is 0.42 kJ/mol per mole of atoms.

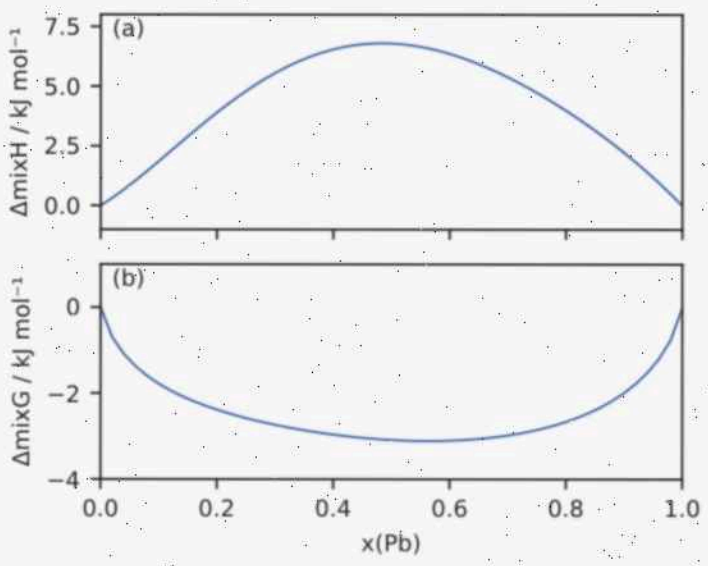


Fig. 2. Calculated (a) mixing enthalpy and (b) mixing Gibbs energy of the liquid at 1450 K.

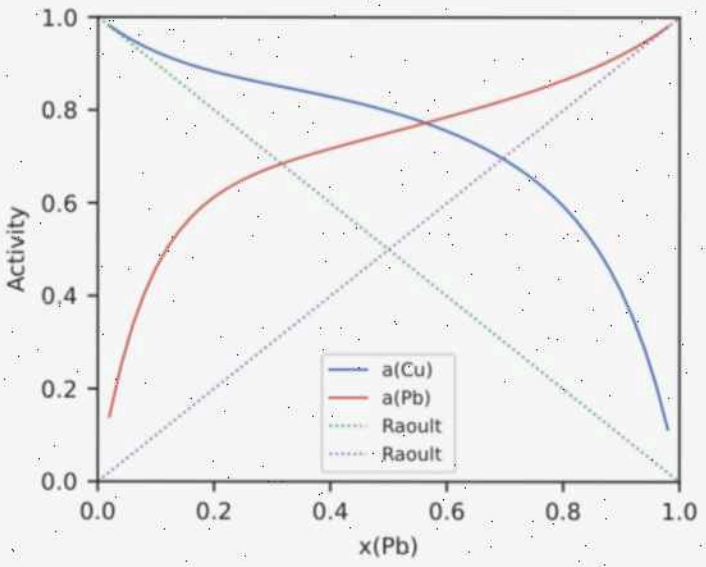


Fig. 3. Calculated activities in the liquid at 1450 K (reference: pure liquids).

Table 2. Calculated formation properties of the intermediate phases (Cu–Pb).

		Calculated at 298.15 K, per mole of atoms	
Phase	**at.% Pb**	**ΔfH / kJ mol⁻¹**	**S / J mol⁻¹ K⁻¹**
HCP_A3	59.0	0.42 ± 2.0	51.15

4. Conclusions

This document is synthetic and exists only to test extraction tools; its truth key lists every boundary, field, invariant, plotted point, table cell and stated value.